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Evaluating zinc bioavailability in dredged sediments of Guanabara bay (BRAZIL): implications for coastal management standards

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Abstract

Guanabara Bay (GB) is an environment contaminated by metals, and it is the bay that receives the most dredging in Brazil and South America. Previous studies have shown the impact of dredging on planktonic and benthic species, which were not correctly measured by the Brazilian dredging standard CONAMA Resolution 454/12 (2012). Given this, the present study compared two acid extractions targeting metal bioavailability (1 M HCl and acetic acid) in sediments from four areas of GB sampled in summer and winter, before and after resuspension. The results showed that Zn concentrations exceeded the standard in all samples in extractions methods and in two seasons after resuspension. In winter, three of the four areas showed concentrations exceeding levels 1 and 2 of CONAMA Resolution 454/12 (2012), both before and after resuspension. The bioavailability change index (BC) was applied to show the remobilization of metals in the sediment after dredging. Results evidenced that Zn remobilization occurred after resuspension for all samples and in most resuspension intervals, indicating an environmental risk related to the resuspension activity. This demonstrates the efficiency of using bioavailable extractions for the monitoring of dredging activities. In addition to being cheaper and easier to perform than the pseudo-total extractions recommended in most global standards, it provides valuable information for improving dredging management policies. According to the results, a revision of the dredging standard is recommended, considering the inclusion of bioavailable extractions.

Keywords Resuspension, Bioavailability change, Acid extraction methods, Estuary

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1 Introduction

Dredging is an indispensable activity for the periodic maintenance of port navigation channels and is used for the recovery and conservation of the flow and depth of silted rivers (Lee et al. 2010; Puig et al. 2022; Wasserman et al., 2013). When dredging is managed correctly, it can have a positive long-term impact on turbidity, sediment balance, and ecology (Sumi & Hirose 2009; Yan & Li 2023). However, dredging remains a controversial activity due to its uncertain negative effects (Yan & Li 2023). Sediment resuspension and removal accelerate the secondary release of contaminants, which directly affects benthic and planktonic organisms (Monte et al. 2025; Yan & Li 2023).

The resuspension of contaminated sediments can increase the bioavailability of metals in the water column that were previously adsorbed to the sediment (Mad-dock et al. 2007; Machado et al. 2011; Monte et al. 2023). Metals do not degrade during dredging; however, they can transform between soluble and insoluble forms (Yan & Li 2023). Although the total concentrations of metals in dredged sediments are reduced, their concentrations increase, mainly in bioavailable fractions (Monte et al. 2025; Yan & Li., 2023; Yu et al. 2019). Another problem related to dredging is the disposal of material in the ocean or in areas within bays, which is the main disposal environment due to space issues on the continent (Chen et al. 2019).

The disposal of this material can affect the environment due to the release of contaminants adsorbed to the sediment, such as metals and organic contaminants, when they come into contact with oxidizing water, increasing the risk to biota (Chen et al. 2019). Dredging directly impacts the physicochemical balance of aquatic systems, increasing siltation and turbidity, consequently reducing light penetration, creating a salinity gradient, and possibly decreasing biomass availability (Soetan et al., 2025).

Furthermore, dredging of contaminated sediments in coastal ecosystems poses a significant risk to biota, as sediment resuspension can lead to the bioaccumulation of trace metals. Additionally, resuspension can expose biota to the risk of poisoning and suffocation (Wasserman et al. 2013; Soetan et al., 2025). However, there remains a gap in global dredging management standards, including Brazilian ones (CONAMA Resolution 454/12 (2012)), which fail to consider metal concentrations before and after resuspension and instead prioritize total or pseudo-total extractions, resulting in inaccurate indicators of ecological risk (Rodrigues et al. 2024; Özseker & Terzi 2025). In this regard, the use of chemical extractions targeting the bioavailable fraction in sediments, such as BCR fraction 1 or 1 M HCl, is essential, enabling better and low-cost assessments, especially for regulatory agencies facing budgetary constraints for acquiring

complex metal quantification technologies (Rauret 1998; Brady et al. 2016; Monte et al. 2025).

The form and occurrence of metals in the environment are important for understanding and predicting contamination and associated ecological risks (Zhong et al. 2011; Brady et al. 2016). There is extensive debate in the literature regarding the best method for assessing the bioavailability of metals in sediments. 1 M HCl extraction is a methodology used to assess bioavailability. Peña-Icart et al. (2014) compared four weak extraction procedures (acetic acid 0.11 M, acetic acid 25%, and 1 M HCl with and without trypsin and pepsin) to evaluate metal bioavailability and concluded that 1 M HCl was the most efficient extraction method to simulate the digestive system of marine organisms.

Another methodology is the use of fraction 1 of the Bureau Communautaire de Référence (BCR) sequential extraction methodology, which is recognized by the European Community Reference Bureau (Rauret 1998). This fraction is related to metals that are exchangeable or associated with carbonate forms and are affected by pH changes (Aguar et al. 2018). However, there is little dissemination in the literature on the use of the two extractions in predicting bioavailability.

In addition to the high cost, most dredging standards worldwide follow European and American standards, which do not reflect the environmental reality of many countries, especially tropical ones (Choueri et al. 2009). Monte et al. (2025) conducted a study on the toxicity of contaminated sediments from GB, analyzing metal concentrations before and after resuspension events and assessing the potentially adverse effects on two species: *Nitokra* sp. and *Echinometra lucunter*, one benthic and the other planktonic, respectively. The results indicated that resuspension significantly affected biota development, corroborating the hypothesis of increased toxicity at the water–sediment interface. During the research, the authors used extraction with 1 M HCl, whose results exceeded the limits established by Brazilian dredging legislation (CONAMA 454/12 (2012)), which is based on the Effect Range Low (ERL) and Effect Range Medium (ERM) values proposed by Long et al. (1995), demonstrating the feasibility of using bioavailability extractions in dredged sediments.

The comprehension and prediction of coastal environment dynamics are fundamental for effective coastal zone management and for interpreting processes such as contaminant dispersion (Reid et al. 2022). In Brazil, dredging activities are recurrent due to the extensive coastline, numerous ports, and silted areas resulting from urbanization and erosion of riverbanks and adjacent areas. In this sense, Guanabara Bay (GB) receives the highest number of dredging operations in Brazil (Valentin et al. 1999; Monte et al. 2023), due to the intense silting of its

hydrographic basin and the maintenance of the navigation channel in port areas. The GB is one of the most metal-polluted aquatic ecosystems in Brazil and worldwide (Abreu et al. 2016; Aguiar et al. 2018). Surrounding the GB watershed is an industrial hub, which accounts for 70% of the industries in the state of Rio de Janeiro and is primarily responsible for the discharge of industrial sewage into this ecosystem (Seixas-Filho et al. 2020).

Several studies have shown that Zn has one of the highest concentrations in sediments in various sections of Guanabara Bay (Abreu et al. 2016; Aguiar et al. 2018). Although zinc is an essential microelement involved in key physiological processes such as metabolism, immune response, growth, and reproduction in marine organisms, excessive concentrations in aquatic ecosystems can lead to bioaccumulation in fish and subsequent human exposure through regular consumption (Asad et al. 2024).

This study is pioneering in integrating data on Zn concentrations before and after sediment resuspension (summer and winter) with the use of two different operationally-defined bioavailability extractions to promote discussion on improving dredging standards worldwide, as previous studies have shown that dredging causes an increase in the concentration of metals in bioavailable fractions. To achieve this objective, chemical and sedimentological analyses were performed to identify and quantify metals after extraction with 1 HCl and 0.11 M acetic acid, respectively. In this sense, new data and previously published data are integrated, allowing the test of following hypotheses: (I) the use of different extractions of the bioavailable fraction may be sensitive to sediment resuspension events; and (II) the different bioavailable fractions can change upon resuspension and reach Zn concentrations above the Brazilian legislation.

2 Materials and methods

2.1 Study area

Guanabara Bay covers an area of 381 km² and has a drainage basin of 4081 km². Approximately 8.6 million people live in 22 municipalities surrounding the bay, located either entirely or partially within the drainage basin (IBGE 2022; Silva et al. 2024). There are several activities in and around the bay, such as ports, shipyards, oil and gas refineries, fishing, tourism, and dredging of the channel and port areas (Aguiar et al. 2022). Sanitary sewage contributes 75% of the organic pollution volume in the bay, and industrial discharges contribute 25% (Silveira et al. 2011). GB is a complex aquatic ecosystem that suffers from severe degradation, including large amounts of metals, due to the discharge of untreated domestic and industrial sewage into the rivers that make up the hydrographic basin (Kjerve et al. 1997; Monte et al. 2025; Rodrigues et al. 2017). Approximately 50% of homes in the urban areas of the drainage basin are connected to the sewage treatment network (Aguiar et al. 2022).

The two port regions of the bay (Rio de Janeiro and Niterói Harbors) (Fig. 1) are contaminated mainly by domestic sewage and metals associated with ships (Aguiar et al. 2018; Baptista Neto et al. 2013). The Iguaçu River estuary is a complex region because it is close to the largest oil refinery in the country and the former Gramacho landfill, which was considered the largest in Latin America until its closure in 2012 (Silveira et al. 2011; Monte et al. 2023).

The Meriti River drains a densely populated region and receives untreated domestic and industrial sewage. In addition, it has long received sewage contaminated with mercury (Hg) and other metals from a chlor-alkali industry located near the riverbanks (Barrocas and Wasserman 1998). These estuaries are located in the

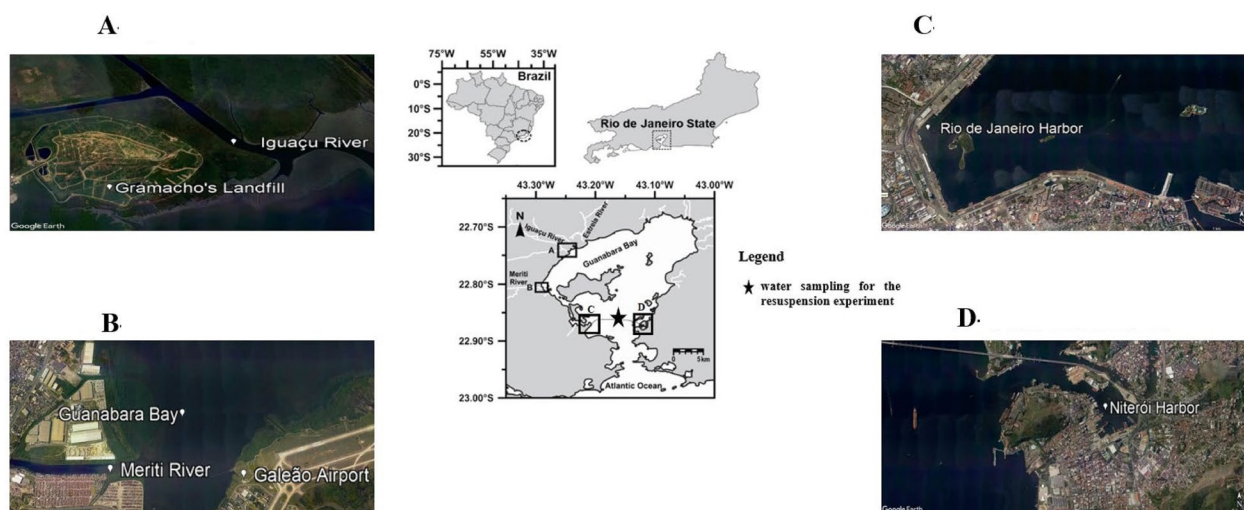


Fig. 1 Study Area – Iguaçu River (A) and Meriti River (B), Rio de Janeiro Harbor (C) and Niterói Harbor (D)

northwestern sector, the most polluted part of the bay, due to the discharge of untreated sewage, which reaches the bay through rivers that drain the state's industrial hub in the metropolitan region of Rio de Janeiro (Kjerve et al. 1997; Baptista-Neto et al. 2013). Another factor contributing to increased pollution in the area is the low water renewal rate due to the hydrodynamic conditions of the sector (Kjerve et al. 1997; Rodrigues et al. 2024).

2.2 Sampling

To test the hypotheses discussed here on different extraction methods, the same samples previously analyzed by Monte et al. (2025) for grain size, organic matter and HCl 1 M extractions for the 1 h and 24 h resuspension times, were used to produce new data on acetic acid extraction in comparison with the HCl 1 M extraction procedure, for an extended number of analyzed resuspension times. Sampling was performed in four areas of Guanabara Bay, divided into two port areas and two estuaries (Rio de Janeiro Harbor, Niterói Harbor, Iguaçu River, and Meriti River) (Fig. 1). In each area, three points were collected in September, 2014 (dry season/winter) and March, 2015 (wet season/summer). The surface sediments (≈ 50 cm to 3 m depth, depending on the area) were obtained with the aid of a deep van Veen search, and the samples were stored in bags and frozen in the laboratory until analyses. Physicochemical parameters (pH, conductivity and temperature) of the surface water were measured in situ using a YSI model 85 multiprobe. This sediment was oven dried for 3 days at moderate temperature (<40 °C) and homogenized for subsequent chemical extraction.

2.3 Resuspension experiments

For the resuspension experiments in the laboratory, 20 g of homogenized wet sediment was weighed in a 125 ml conical flask previously decontaminated according to Morse (1994); Machado et al. (2011) and Monte et al. (2025), and 100 ml of unfiltered water collected from the Guanabara Bay channel area was added. The use of the cited amounts of sediment and water came from an adaptation of the study by Morse (1994) that gave rise to this study and others performed by the same group (Machado et al. 2011). During agitation, the flasks were kept open, in contact with the atmosphere, using a mechanical shaker at a speed of 200 rpm. The agitation speed used in the experiments was within the range reported in the literature (Acquavita et al. 2012; Fetters et al. 2016; Monte et al. 2025; Xie et al. 2019;).

The resuspension test was performed at six-time intervals: 30 min (T1), 1 h (T2), 3 h (T3), 6 h (T4), 12 h (T5) and 24 h (T6). The time interval T0 corresponds to the sediment before the resuspension. The choice of the intervals was based on previous studies which showed that it is during the first hour that changes in

bioavailability occur (Machado et al. 2011; Monte et al. 2025; Morse 1994). The 24-h interval was included as in other studies (Morse 1994), to compare the stability of contaminants over shorter intervals and simulate longer-duration resuspension events. All samplings were conducted in duplicates.

After the resuspension tests, physicochemical parameters (pH, conductivity and temperature) of the water were measured using a YSI model 85 multiprobe, after the water was filtered and the sediment was separated by centrifugation. This sediment was oven dried for 3 days at moderate temperature (<40 °C) and homogenized for subsequent chemical extraction.

2.4 Grain size and organic matter

The grain size characterization was performed with 1.5 g of wet sediment sample placed in a 50 mL falcon tube, and 40 mL of hexametaphosphate dispersing agent (40 g/L) was added, followed by stirring for 24 h. The particle size determination was performed in a laser particle analyzer (CILAS Shimadzu 1064). The results were obtained with the aid of the GRADSTAT version 8 program.

Organic matter (OM) was characterized by loss on ignition (LOI) which used 1 g of sample and submitted to the muffle furnace for 4.5 h at a temperature of 450 °C. The OM was obtained by comparing the mass values before and after calcination (Heiri et al. 2001; Morelle et al. 2020).

2.5 Metals determinations

The choice of Zn as the only metal in this study is related to the high concentrations of the metal in GB sediment due to domestic and industrial sewage, port activity, and because it is an essential microelement for aquatic organisms. The reactive phase (i.e., adsorbed and/or associated with carbonates, monosulfides, Fe and Mn oxides hydroxides and clay minerals) was extracted by 16 h agitation in 1 M HCl solution and sediment samples (Morse et al. 1994; Larner et al. 2008; Leleyter et al. 2012; Wu et al. 2025). It is a common approach for evaluating metal bioavailability (Huerta-Diaz and Morse, 1990; Machado et al. 2011; Morse 1994; Peña-Icart 2014; Wu et al. 2025).

In this procedure, 40 mL of 1 M HCl solution was added to a polypropylene tube containing dried sediment (7.5 g). The tubes were agitated for 16 h. After this period the samples were centrifuged (3,000 rpm, for 5 min), the supernatant was separated for the determination of Zn concentrations by inductively coupled plasma optical emission spectrometry (ICP-OES), using Jobin–Yvon model Ultima 2 sequential ICP OES Spectrometer Optic system. The detection limits were: 0.0004 mg L⁻¹. PACS 3-certified material was used, and the recovery ranged from 96 to 103%.

2.6 Sequential extraction of metals (BCR) (only step1)

Methods that utilize geochemical fractionation can provide information into the transport and transformation mechanism of elements under various environmental conditions (McAlister et al., 2010). One step of the sequential extraction procedure proposed by the Community Bureau of Reference (BCR) (Sutherland, 2010). The method is composed of three geochemical fractions, whose analytical procedures were performed as described in the step presented below:

Fraction 1 (F1) consists of exchangeable ions and/or associated with carbonates. We added to the centrifuge tubes 0.5 g of dry and homogenized sediment and 20 mL of acetic acid 0.11 M. The mixture was inserted in ultrasound for 30 min. After this step, the sample was centrifuged at 3,000 rpm for 20 min. The resulting supernatant was removed and stored in a 50 mL plastic bottle and extract was analyzed. The residue was then washed, in the centrifuge flask, with 20 mL of deionized water and 5 min and discard.

2.7 Bioavailability Changes (BC)

The Bioavailability Change Index (BCI) developed by Monte et al. (2015) was adapted by Rodrigues et al. (2017) and included only the concentrations of metals in the reactive phase (BC). Then, a simple calculation of reactive trace metal losses and gains as a percentage related to T0 (in natura samples, before resuspension) was performed, as explained in the following Eq. 1:

$$BC = (([MeAR] - [MeBR]) / [MeBR]) * 100 \quad (1)$$

where [MeAR] is the metal concentration in the reactive phase after resuspension and [MeBR] is the metal concentration in the reactive phase before resuspension. According to Monte et al., (2019), BC could be negative or positive, which means that bioavailability could be reduced or increased (respectively) in surface sediments. In this study, the BC was calculated for the two operationally-defined extractions separately. The scale used to classify the samples reflects the intensity of the changes in the sediments after resuspension:

Increase on Bioavailability/ Decrease on Bioavailability
 0 - 15 % = none or low 0 - -15 % = none or low
 15 - 50 % = moderate -15 - -50 % = moderate
 50 - 100 % = considerable -50 - -100 % = considerable
 > 100 % = very high > 100 % = very high

2.8 Data analysis

Statistical analyses were performed using the softwares PAST 3 and STATISTICA 8.0. After applying the Shapiro-Wilk normality test. All data was normal. Pearson correlation and Student's t-test were tested to verify

possible relationships between OM, Fe, Mn and Zn of two seasons. Principal Component Analysis (PCA) test was applied to verify possible relationships between Zn concentration in two extractions, pH, EC, temperature and organic matter content. Such multivariate analysis allows investigations with a large amount of data (Li et al. 2019).

3 Results

3.1 Winter

Niterói Harbor (NH), Iguaçu River (IR), Meriti River (MR), and Rio de Janeiro Harbor (RJH) showed a tendency for temperature and EC to increase after resuspension compared to T0 (Table 1), while pH results showed a tendency to decrease after resuspension. In terms of grain size, NH, IR, and MR were composed of 100% fine fraction (silt + clay), whereas RJH showed a small percentage of sand in its composition (Fig. 2). However, it was composed mainly (more than 80%) of fine fraction.

Regarding organic matter (OM), all samples before and after resuspension showed results above 10% (Fig. 3). NH and MR exhibited an oscillatory trend before and after resuspension. IR and RJH showed increasing trends after resuspension. The concentrations of Fe, Mn (Fig. 4), and Zn (Fig. 5) in the 1 M HCl extraction showed an increasing trend at all points after resuspension. Zn concentrations were above the ERL for all resuspension intervals in the NH and MR (Fig. 5). However, RJH showed all intervals after resuspension above the ERM, indicating a high risk to biota.

The iron concentrations in the acetic acid extraction showed an upward trend at all points after resuspension (Fig. 4). Mn concentrations showed an upward trend in the NH, IR, and RJH groups after resuspension. Meanwhile, MR showed an oscillatory trend after resuspension. The Zn content tended to increase after resuspension in all samples. Zn concentrations were above the ERL at all intervals after resuspension in the NH, MR, and RJH groups (Fig. 5).

The statistics for 1 M HCl extraction showed a positive correlation between Zn and Mn ($r=0.96$; $p<0.05$) only in NH. Student's t-test was applied between metal concentrations and OM in winter and summer for each point before and after resuspension. IR and RJH showed statistical differences for Zn ($t=20.4$; $p\leq 0.05$) and ($t=14.8$; $p\leq 0.05$), respectively. This may indicate different processes and sources.

Regarding the statistics of acetic acid extraction, NH showed a positive correlation between Zn and Mn ($r=0.80$, $p<0.05$). MR showed a positive correlation between Zn and OM ($r=0.67$; $p<0.05$) and Zn and Fe ($r=0.90$; $p<0.05$). RJH showed a positive correlation between Zn and Fe ($r=0.68$; $p<0.05$) and Zn and Mn ($r=0.66$; $p<0.05$). The application of Student's t-test

Table 1 Physical–chemical parameters of Niterói Harbor (NH), Iguaçu River (IR), Meriti River (MR) and Rio de Janeiro Harbor (RJH) before and after resuspension in the summer

Season	Samples	Intervals	EC ($\mu\text{S}/\text{cm}$)	Temperature ($^{\circ}\text{C}$)	pH
Winter	NH	T0	74.9	21.2	8.0
		T1	75.8 $\pm$ 0.3	23.9 $\pm$ 0.1	8.3 $\pm$ 0.0
		T2	76.3 $\pm$ 0.5	24.1 $\pm$ 0.1	8.1 $\pm$ 0.2
		T3	77.7 $\pm$ 0.4	24.1 $\pm$ 0.1	7.5 $\pm$ 0.1
		T4	78.3 $\pm$ 1.1	25.0 $\pm$ 0.3	7.5 $\pm$ 0.1
		T5	78.9 $\pm$ 0.8	25.7 $\pm$ 0.3	7.5 $\pm$ 0.0
	MR	T6	78.5 $\pm$ 1.1	24.8 $\pm$ 0.3	7.9 $\pm$ 0.4
		T0	74.9	21.2	8.0
		T1	73.7 $\pm$ 1.5	24.3 $\pm$ 0.9	8.3 $\pm$ 0.2
		T2	74.3 $\pm$ 0.3	24.4 $\pm$ 0.8	8.3 $\pm$ 0.1
		T3	73.9 $\pm$ 0.5	24.1 $\pm$ 0.2	8 $\pm$ 0.1
		T4	71 $\pm$ 1.6	25.3 $\pm$ 0.4	7.3 $\pm$ 0.3
	IR	T5	75.1 $\pm$ 0.4	25.5 $\pm$ 0.1	6.5 $\pm$ 1.1
		T6	75.1 $\pm$ 0.7	24.8 $\pm$ 0.2	8 $\pm$ 0.4
		T0	74.9	21.2	8.0
		T1	72.1 $\pm$ 1	23.8 $\pm$ 0.1	8.2 $\pm$ 0.1
		T2	71.3 $\pm$ 0.9	24.1 $\pm$ 0.2	8.1 $\pm$ 0.1
		T3	71.3 $\pm$ 0.9	23.9 $\pm$ 0.4	7.9 $\pm$ 0.1
	RJH	T4	70.9 $\pm$ 0.2	24.9 $\pm$ 0.2	7.8 $\pm$ 0.1
		T5	70.9 $\pm$ 0.2	25.2 $\pm$ 0.3	7.3 $\pm$ 0.6
		T6	73 $\pm$ 1.8	24.7 $\pm$ 0.14	7.8 $\pm$ 0.2
		T0	74.9	21.2	8.0
		T1	75.4 $\pm$ 0.4	24 $\pm$ 0.1	7.9 $\pm$ 0.3
		T2	76 $\pm$ 0.5	24.2 $\pm$ 0.2	7.6 $\pm$ 0.42
Summer	NH	T3	76.2 $\pm$ 0.5	24.6 $\pm$ 0.2	7.7 $\pm$ 0.09
		T4	76.3 $\pm$ 0.9	25.2 $\pm$ 0.2	7.4 $\pm$ 0.2
		T5	78.1 $\pm$ 0.5	25.9 $\pm$ 0.1	6.7 $\pm$ 0.5
		T6	77.3 $\pm$ 1.4	25.0 $\pm$ 0.4	7.1 $\pm$ 0.4
		T0	76.1	26.1	8.5
		T1	71.9 $\pm$ 0.64	26.7 $\pm$ 0.42	8.6 $\pm$ 0.04
	MR	T2	70.9 $\pm$ 0.12	24.6 $\pm$ 0.12	7.8 $\pm$ 0.09
		T3	71.2 $\pm$ 0.3	25.9 $\pm$ 0.1	7.7 $\pm$ 0.08
		T4	71.5 $\pm$ 0.59	26.2 $\pm$ 0.08	7.5 $\pm$ 0.15
		T5	72.1 $\pm$ 0.52	25.2 $\pm$ 0.15	7.3 $\pm$ 0.13
		T6	74.1 $\pm$ 0.35	25.8 $\pm$ 0.2	7.6 $\pm$ 0.43
		T0	76.1	26.1	8.5
	IR	T1	68.2 $\pm$ 0.39	26.4 $\pm$ 0.12	8.4 $\pm$ 0.16
		T2	67.2 $\pm$ 0.85	24.7 $\pm$ 0.08	7.9 $\pm$ 0.14
		T3	67.1 $\pm$ 0.93	26.0 $\pm$ 0.08	7.9 $\pm$ 0.08
		T4	66.6 $\pm$ 1.03	26.0 $\pm$ 0.12	7.9 $\pm$ 0.10
		T5	67.2 $\pm$ 0.40	24.9 $\pm$ 0.12	7.1 $\pm$ 0.10
		T6	70.0 $\pm$ 2.23	25.9 $\pm$ 0.20	7.5 $\pm$ 0.14
	RJH	T0	76.1	26.1	8.5
		T1	67.4 $\pm$ 0.57	26.5 $\pm$ 0.33	8.2 $\pm$ 0.09
		T2	65.2 $\pm$ 0.88	24.8 $\pm$ 0.08	7.8 $\pm$ 0.02
		T3	64.7 $\pm$ 0.82	25.8 $\pm$ 0.05	7.8 $\pm$ 0.06
		T4	64.6 $\pm$ 1.21	26.0 $\pm$ 0.16	7.6 $\pm$ 0.22
		T5	65.5 $\pm$ 1.38	25.1 $\pm$ 0.08	6.7 $\pm$ 0.69
	RJH	T6	67.7 $\pm$ 0.68	25.6 $\pm$ 0.08	7.3 $\pm$ 0.87
		T0	76.1	26.1	8.5
		T1	71.4 $\pm$ 0.32	26.4 $\pm$ 0.15	8.6 $\pm$ 0.07
		T2	70.9 $\pm$ 0.2	24.8 $\pm$ 0.11	8.1 $\pm$ 0.06
		T3	71.6 $\pm$ 0.3	25.9 $\pm$ 0.13	8.0 $\pm$ 0.10
		T4	71.0 $\pm$ 0.66	26.1 $\pm$ 0.23	7.8 $\pm$ 0.07
		T5	71.4 $\pm$ 0.49	25.3 $\pm$ 0.08	7.5 $\pm$ 0.04
		T6	73.2 $\pm$ 1.42	26.0 $\pm$ 0.33	7.4 $\pm$ 0.08

*The data for the summer campaign was published in Monte et al. 2023

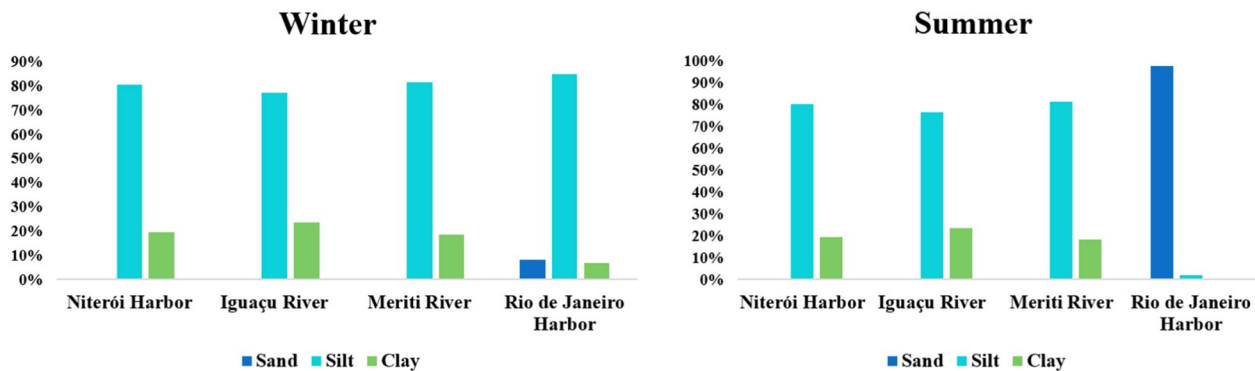


Fig. 2 Grain size (%)

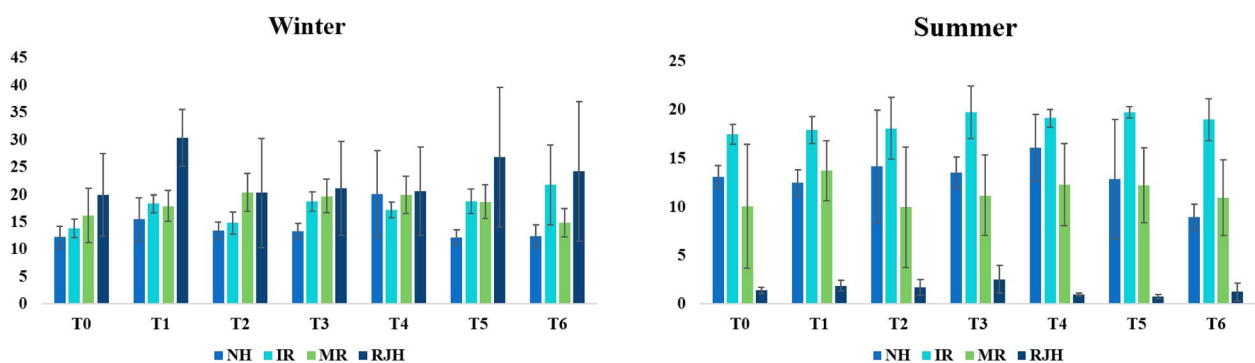


Fig. 3 Organic matter contents before and after resuspension in Niterói Harbor (NH), Iguaçu River (IR), Meriti River (MR) and Rio de Janeiro Harbor (RJH) (%)

showed statistical differences for Zn in NH, IR, and RJH ($t=11.7$; $p \leq 0.05$), ($t=35$; $p \leq 0.05$), and ($t=2.4$; $p \leq 0.05$), respectively. Organic matter also showed a statistical difference between winter and summer in MR and RJH ($t=7.3$; $p \leq 0.05$) and RJH ($t=14.3$; $p \leq 0.05$).

In NH and RJH, PCA analyses showed positive correlations between Zn concentrations in both extractions, EC and temperature (Fig. 6). Meanwhile, pH showed an opposite trend to Zn concentrations (both extractions), a similar behavior observed in IR and MR. Zn concentrations (both extractions) were correlated with OM and temperature in MR and IR.

The BC results showed 24 positive results and five negative result for 1 M HCl extraction (Table 2), suggesting that Zn was remobilized after resuspension at all intervals. The BC results for acetic acid extraction showed 19 significant results, 18 of which were positive, indicating remobilization after resuspension, and 1 was negative (Table 2).

3.2 Summer

The physicochemical parameters showed similar behavior at all points. EC, temperature, and pH showed a decreasing trend after resuspension in the NH, IR, MR, and RJH (Table 1). The grain size was predominantly fine

(above 80%) in the NH, IR, and MR seasons (Fig. 2). However, RJH had a predominantly sandy percentage (above 80%) (Fig. 2). The OM percentage showed a decreasing trend after resuspension in the NH (Fig. 3), whereas the IR showed an increasing trend after resuspension. The MR and RJH fluctuated, showing no definite trend after resuspension.

The Fe concentrations in the 1 M HCl extraction tended to increase after resuspension in NH, IR, and RJH. MR did not show a trend, with concentrations fluctuating after resuspension (Fig. 4). Mn concentrations fluctuated after resuspension at all points. The Zn content fluctuated after resuspension, showing no clear trend for NH, IR, and RJH (Fig. 7). The MR showed an increasing trend after resuspension. Zn concentrations were above the ERL in the NH and IR in all resuspension intervals, while the MR showed three results above the ERL and three above the ERM after resuspension (Fig. 7).

The Fe concentrations in the acetic acid extraction showed a downward trend after resuspension at all points (Fig. 4). Mn concentrations showed a downward trend after resuspension in the NH, IR, and MR, whereas the RJH showed an oscillatory trend. Zn showed an increasing trend in concentrations in MR and RJH after resuspension, while NH and IR oscillated in concentration

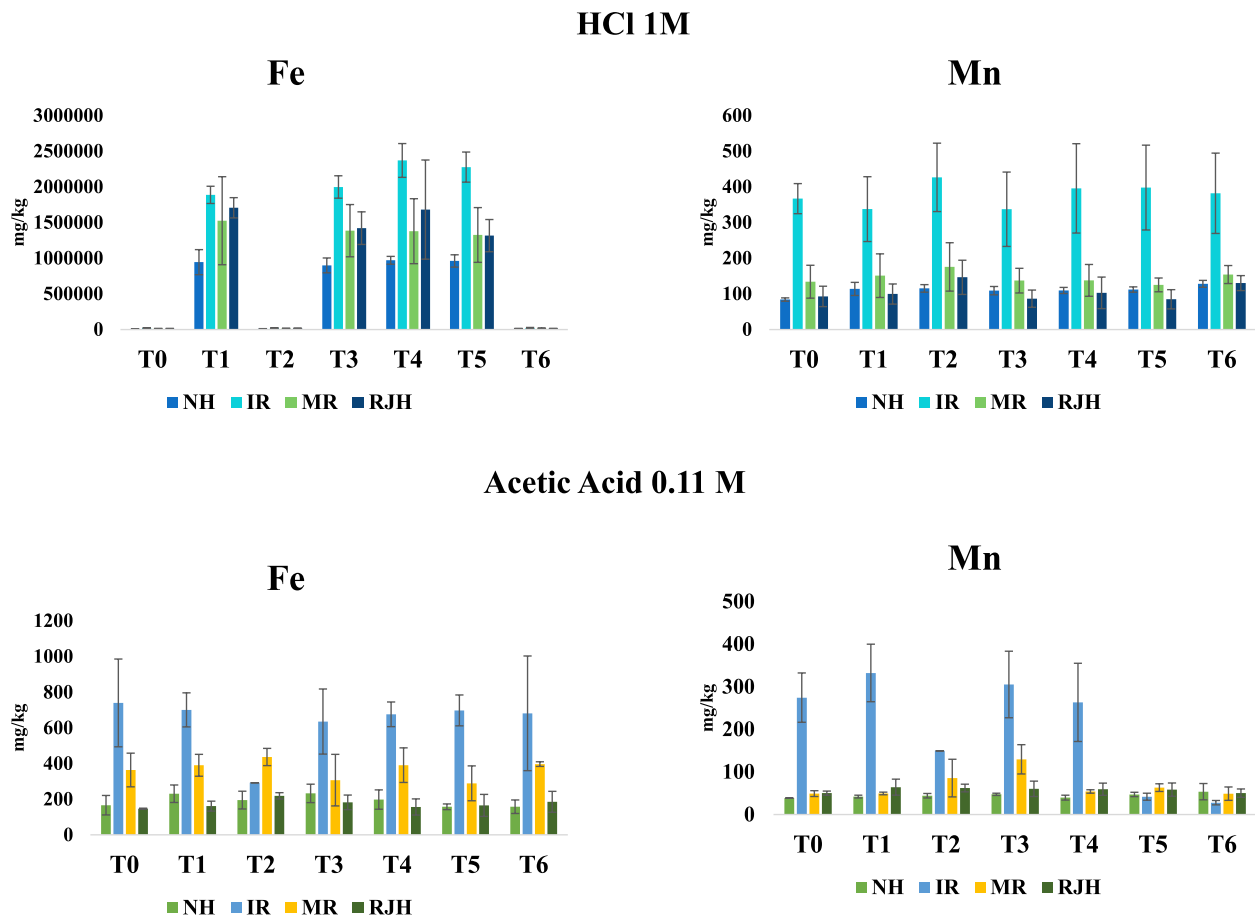


Fig. 4 Fe and Mn concentrations before and after resuspension in Niterói Harbor (NH), Meriti River (MR), Iguaçu River (IR) and Rio de Janeiro Harbor (RJH) in the winter (mg/kg)

after resuspension. NH and IR showed concentrations above the ERL after resuspension. MR showed four concentrations above the ERM and two above the ERL (Fig. 7), suggesting a risk to biota after resuspension.

In relation to statistics, NH showed a positive correlation between Zn and Fe, Zn and Mn, Zn and OM, and OM and Mn ($r=0.76$; $p<0.05$), ($r=0.77$; $p<0.05$), ($r=0.78$; $p<0.05$), ($r=0.68$; $p<0.05$), respectively. MR showed positive correlations between Zn and Fe, Zn and Mn, Fe and Mn, Fe and OM, and Mn and OM ($r=0.70$; $p<0.05$), ($r=0.68$; $p<0.05$), ($r=0.97$; $p<0.05$), ($r=0.71$; $p<0.05$), ($r=0.74$; $p<0.05$), respectively. RJH showed a positive correlation between Fe and Mn ($r=0.86$; $p<0.05$).

The correlations related to acetic acid extraction in NH were positive between Zn and Fe ($r=0.74$; $p<0.05$). In IR, there was a positive correlation between Zn and Fe; Zn and Mn; and Fe and Mn ($r=0.92$; $p<0.05$), ($r=0.83$; $p<0.05$), ($r=0.76$; $p<0.05$). MR showed a positive correlation between Zn and Fe, and Zn and OM ($r=0.83$; $p<0.05$), ($r=0.71$; $p<0.05$), respectively, and a negative correlation between Fe and OM ($r=-0.61$; $p<0.05$).

In the NH and RJH, PCA analyses showed positive correlations between Zn concentrations in both extractions, temperature, and EC (Fig. 6). While pH showed an opposite trend to Zn concentrations (both extractions), a similar behavior was observed in winter. MR and IR showed similar behavior to winter, with Zn concentrations (both extractions) correlated with OM and temperature, whereas pH and EC showed the opposite behavior.

4 Discussion

Estuarine areas with fine sediments can function as active colloids for the adsorption and complexation of metal ions (Shao et al. 2020). In addition, metals can be retained in sediment due to coprecipitation with oxides, cation exchange, and complexation with OM in coastal areas (Silva et al. 2024). In this study, the samples, mainly in winter, had fine grain size, which may explain the concentrations above the ERL and ERM in the sediment. Furthermore, finer grain sizes provide a larger surface area, higher surface charge, and greater complexed organic matter (OM) content, which may explain the

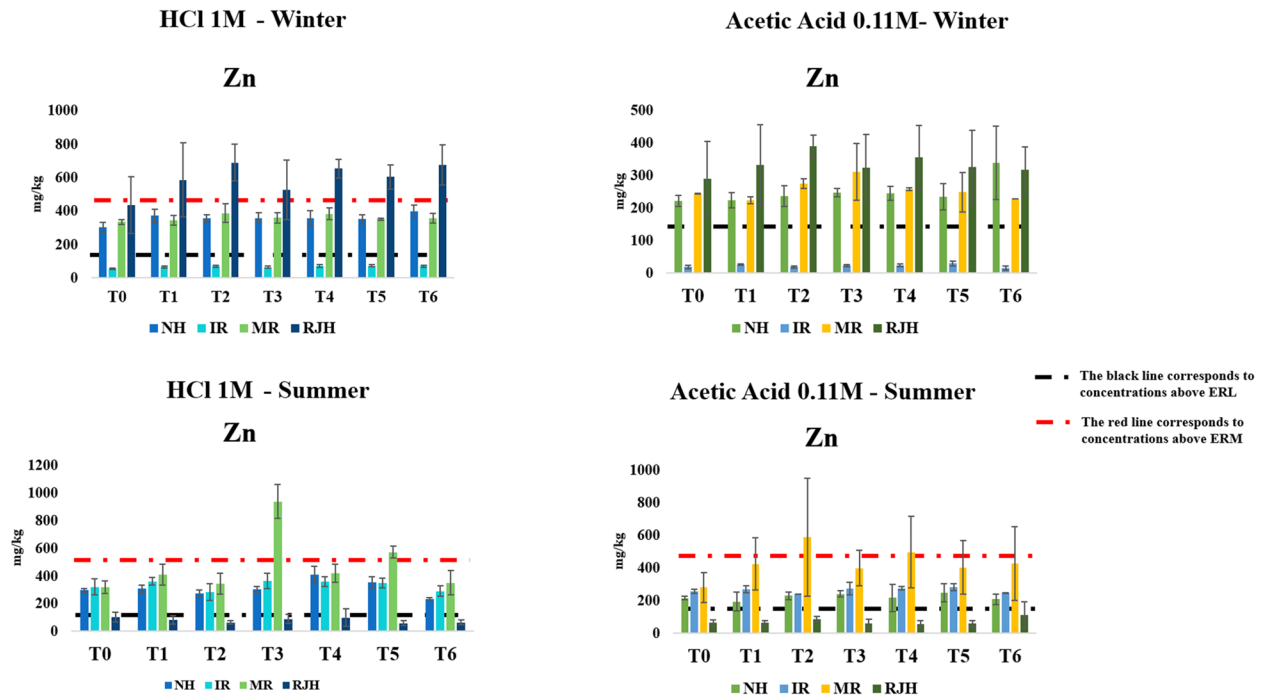


Fig. 5 Zn concentrations before and after resuspension in Niterói Harbor (NH), Meriti River (MR), Iguaçu River (IR) and Rio de Janeiro Harbor (RJH) in the winter and summer (mg/kg)

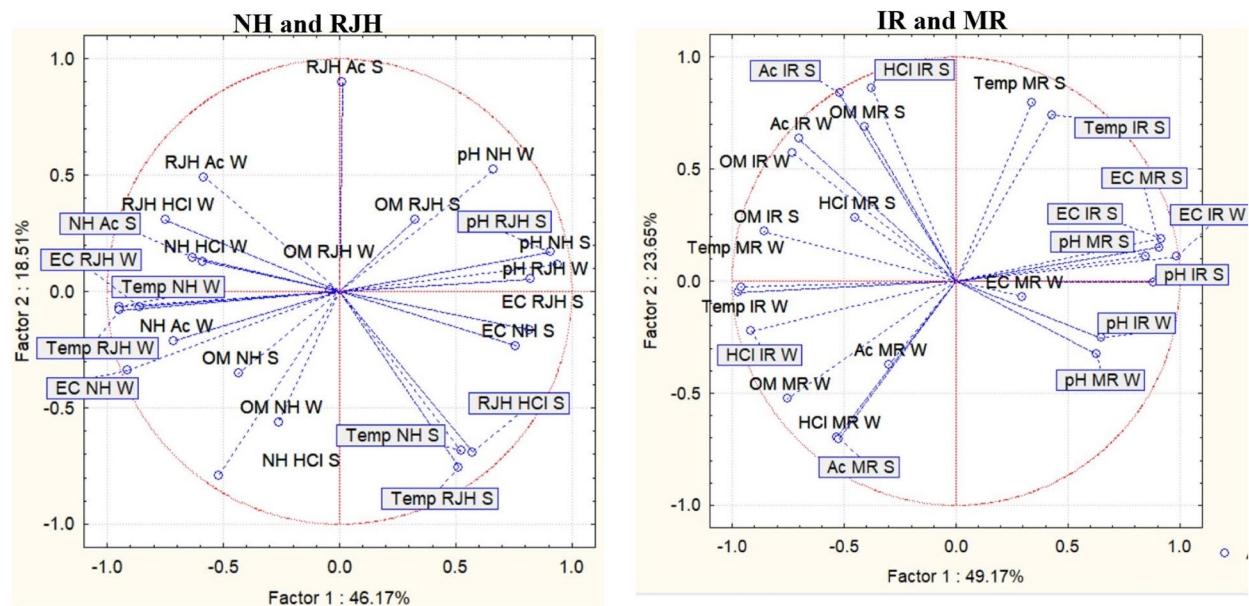


Fig. 6 PCA analysis. The plot of before and after resuspension in winter (W) and summer (S) using the physicochemical parameters, the Zn concentration, of organic matter (OM) in Niterói Harbor (NH), Rio de Janeiro Harbor (RJH), Iguaçu River (IR) and Meriti River (MR)

elevated metal concentrations found in fine-grained sediments (Özseker et al. 2022a).

In addition, the highest concentrations of metals were found in the winter campaign, which may be related to the dry season in Rio de Janeiro, with less water input into the system. The lower concentrations of metals before and after resuspension in summer may be related

to the dilution effect due to rainfall, which consequently increases the flow of rivers in the watershed (Monte et al. 2025).

In summer, there was a greater deposition of sandy sediments and fluvial OM. RJH presented a sandy grain size in summer, which may be related to the entry of swell waves in the region, and consequently directly impacts

Table 2 Bioavailability Changes (BC) of Zn in HCl 1M and acetic acid 0.11M in winter and summer for Niteroi Harbor (NH), Iguaçu River (IR), Meriti River (MR) and Rio de Janeiro Harbor (RJH)*

Intervals		1M HCl				0.11M Acetic Acid			
		NH	IR	MR	RJH	NH	IR	MR	RJH
Winter	T1	22.8	16.3	3.5	34.8	0.4	37.0	-8.0	15.2
	T2	17.2	24.6	15.9	58.4	6.3	-0.4	12.8	34.6
	T3	18.1	14.7	7.5	21.1	11.5	24.0	27.9	11.8
	T4	17.4	28.2	14.6	50.2	10.4	29.9	5.9	22.8
	T5	16.1	29.4	5.1	38.9	5.4	58.6	2.1	12.5
	T6	32.1	24.5	6.7	55.3	52.6	-18.8	-6.3	10.1
Intervals		1M HCl				0.11M Acetic Acid			
		NH	IR	MR	RJH	NH	IR	MR	RJH
Summer	T1	3.2	13.1	28.2	-18.5	-9.5	5.3	50.8	-2.4
	T2	-7.8	-10.7	7.0	-38.5	6.6	-6.5	109.4	34.7
	T3	2.4	13.8	193.7	-12.9	12.3	6.5	41.1	-3.8
	T4	38.6	12.5	31.1	-2.5	1.2	7.2	76.6	-9.4
	T5	19.1	9.2	78.9	-42.3	15.1	9.5	42.9	-7.6
	T6	-21.8	-9.2	9.5	-37.3	-3.2	-4.1	51.3	76.7

*Results above 15 are in bold

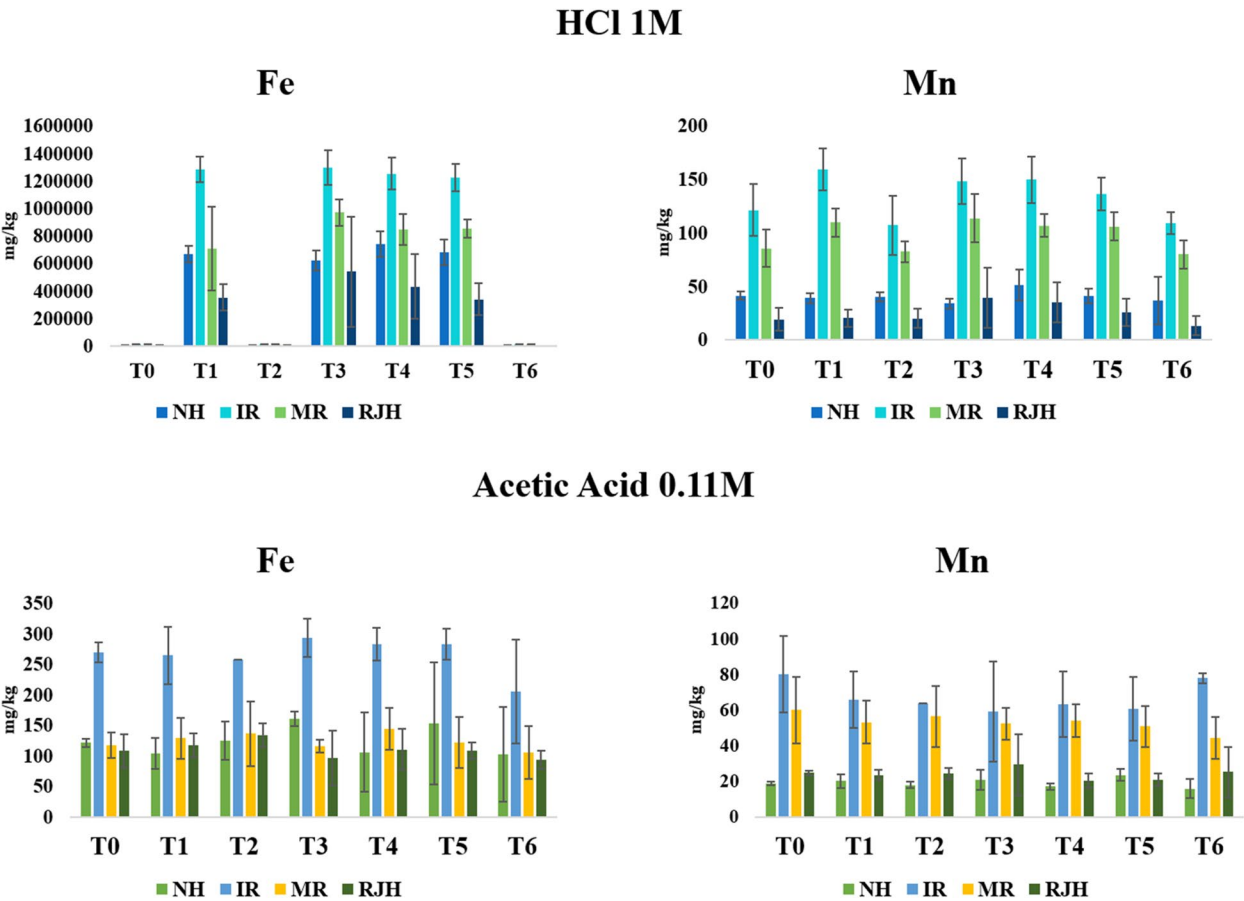


Fig. 7 Fe and Mn concentrations before and after resuspension in Niterói Harbor (NH), Meriti River (MR), Iguaçu River (IR) and Rio de Janeiro Harbor (RJH) in the summer (mg/kg)

metal concentrations (Baptista-Neto et al. 2013; Monte et al. 2025).

The higher concentrations of metals after resuspension at some intervals in both campaigns may be related to the lower probability of vertical movement of metals in the sediment profile, it is likely that metals move from the aquatic phase to the solid phase if changes in the chemical stability occur (Shao et al. 2020; Yang et al. 2014). However, the transfer of metals from the aquatic phase to the solid phase depends on the variation in factors of influence (pH, Eh, salinity, temperature, etc.), atomic properties, and chemical speciation of the metal (Shao et al. 2020; Yang et al. 2014).

This behavior was also observed by Monte et al. (2025), who reported a transfer from the aquatic to the solid phase during winter, and the opposite trend in summer, after the resuspension of sediments. Sediment resuspension can cause a series of physicochemical changes, such as pH and redox state, which induce the remobilization of elements associated with sediment particles, such as metals, especially in polluted regions (Champiwat et al. 2023; Feters et al. 2016).

The Fraction 1 of the BCR is related to metals bioavailable for uptake by benthic fauna and is primarily bound to carbonates. In this sense, these metals are susceptible to be released into the water column during dredging operations (Di Nanno et al. 2007; Vicente et al., 2024).

When comparing the extractions of 1 M HCl and 0.11 M acetic acid, all metals in this study showed higher concentrations in 1 M HCl, which has a high capacity to dissolve carbonate, Fe and Mn oxides, and part of the combined organic and sulfurous metals. However, it cannot dissolve metals in their residual form (Leleyter et al. 2012; Yu et al. 2021).

According to Yu et al. (2021), who studied the bioavailability of metals in contaminated sediments, Cd and Pb had lower concentrations in 1 M HCl extraction than in BCR extraction. The authors attributed this difference to several factors, such as the possibility of complex formation, such as CaCl_2 , due to Fe and Ca competing with other metals to consume the HCl solution and polluted environments. Cd and Pb showed higher concentrations in the 1 M HCl extractions in less polluted environments. However, in this study, Fe, Mn, and Zn presented higher concentrations in 1 M HCl. GB is an environment polluted by metals, mainly Zn in many regions of the bay (Aguiar et al. 2018; Machado et al. 2011; Monte et al. 2025).

The Zn concentration in GB is related to industrialization in the surrounding area and the release of domestic sewage (Monte et al. 2025; Silveira et al. 2011). The high concentrations of Zn in this study before resuspension, mainly in port areas, may be related to the washing of ship holds and the natural wear and tear of paint on the

hulls of ships docked at the port (Bejarano-Ramirez et al. 2017; Correa et al. 2023). Zn has a high affinity for OM, Fe, and Mn oxides and hydroxides in aquatic environments (Barreira et al. 2025; Monte et al. 2025). However, the oxidation of OM and sulfides after resuspension may explain the remobilization of metals in this study, especially in winter, increasing the risk to biota (Caetano et al. 2003; Monte et al. 2025).

In this study, Zn showed a positive correlation with Mn in both extractions. According to Corrêa et al. (2023), in a study on contaminated sediments in ports on the Equatorial Margin in Brazil, Mn is a carrier of metals to sediments in estuarine areas, which may be related to its high resistance to flocculation, remaining longer in the water column and, consequently, adsorbing a higher metal content than Fe. However, Mn tends to form oxyhydroxide when oxygen-rich seawater enters the estuary (Corrêa et al. 2023), which explains the results of this study after resuspension in oxidizing water.

The application of BC in both extractions showed the remobilization of Zn after resuspension, demonstrating the efficiency of using bioavailable extractions as a predictive tool in the management of contaminated dredged sediments from highly polluted areas. Monte et al. (2015) conducted a study on dredged sediments from a mesotrophic estuary using the sequential extraction methodology of 1 M HCl and HNO_3 . The authors showed that a large percentage of Zn was in the bioavailable fraction and that part of it was remobilized after the application of the BCI index.

During sediment resuspension events, metals can become available in the water column, but this availability may be short-lived (Maddock et al. 2007). However, metals newly associated with the solid phase may become bioavailable after resuspension due to adsorption onto particle surfaces (Caetano et al. 2003; Monte et al. 2015). Most of the results in this study were above the current standard, which demonstrates a risk to biota, as observed in other studies that used bioassays with dredged sediments (Monte et al. 2025).

Furthermore, the chemical criteria for sediments established by CONAMA Resolution 454/12 (2012) are derived from ecotoxicological values adopted by environmental agencies in the United States and Canada. However, despite concerns regarding the protection of benthic organisms, these Brazilian Sediment Quality Guidelines are based on international criteria that do not align with the environmental reality of tropical regions, particularly in Brazil. The lack of site-specific data on sediment quality in tropical countries, combined with the significant diversity of their coastal environments, renders the currently adopted standards ineffective for accurately assessing risks to biota (Moreira et al. 2021).

Given the physicochemical complexity of sediments in dredged aquatic systems, the establishment of coherent and robust environmental policies is imperative (Lee et al. 2010). Nonetheless, existing legal frameworks are often fragmented and insufficient to address the multifaceted ecological risks associated with dredging activities, highlighting the need for more stringent regulations and standardization (Safhi et al. 2024). The adoption of standardized assessment plays a critical role in supporting integrated monitoring efforts grounded in multidisciplinary approaches, thereby facilitating the effective communication of environmental risks to both decision-makers and the broader public (Linkov et al. 2009; d'Errico et al. 2021). Advancing this agenda depends on sustained legislative action and active societal engagement (Riza et al. 2023).

The results of the present study showed the feasibility and highlight the use of bioavailable extractions as tools to assess the contamination of dredged sediments, and the possible revision of current dredging standards in Brazil and worldwide, which mostly use pseudo-total extractions. The use of pseudo-total extraction does not demonstrate the environmental reality of areas subjected to dredging. In addition, the standards do not use laboratory resuspension tests, and other studies have shown that there is remobilization of metals in the sediment and water column. Future studies may create new standard limits, especially in tropical countries, which do not reflect the environmental reality when limits from temperate countries are used.

5 Conclusion

Our results showed that the use of bioavailable extractions was effective, especially for extraction in 1 M HCl, which proved to be more efficient than extraction with acetic acid. This can be adopted as a metal extraction method for dredging standards, better reflecting the risks to aquatic biota and being a cheaper methodology than the acid extractions used in most standards in other countries. Contrary to expectations, metal concentrations were higher in winter due to factors such as rainfall availability, and hydrodynamics which differ between summer and winter and affect the bioavailability of metals.

Most of the results exceeded the ERL and ERM limits and, consequently, the limits established by Brazilian dredging regulations, indicating that the risks to biota following resuspension are high. It is necessary to evaluate in greater detail the inclusion of resuspension tests in dredging regulations in accordance with each country's specifications. Furthermore, the management of dredged sediments must be reviewed by adopting criteria that better reflect the environmental realities of each country.

It is worth noting that tropical countries are not adequately addressed by existing dredging regulations, as current standards do not consider physical–chemical criteria or bacterial activity that differ from those in temperate regions; they merely replicate values and conditions from the countries that conducted the initial studies and established the standards. Therefore, we recommend the incorporation of bioavailability-based extraction methods into Brazilian dredging legislation and a revision of the limits of the ERL and the ERM tropical countries.

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Authors' contributions

All authors: Analysis, sample preparation, experimental design, and data interpretation. CNM, APCR, MCS, LIDS, RCC, and WM: Writing – original draft and editing, data curation, and investigation. CNM, APCR, GM, LIDS, HS, AN, and LL: Analyses. CNM, APCR, MCS, WS, and WM: conceptualized, supervised, edited, and reviewed the manuscript. All authors edited and reviewed the manuscript.

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Data availability

The datasets generated and analyzed during the current study are available from the corresponding author upon request.

Declarations

Ethics approval and consent to participate

Not applicable. This study did not involve human participants, animals, or sensitive personal data.

Consent for publication

Not applicable. The manuscript does not contain individual-level data or images requiring consent for publication.

Competing interests

The author declares no competing interests.

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References

- Abreu IM, Cordeiro RC, Soares-Gomes A, Abessa DMS, Maranhão LA, Santelli RE (2016) Ecological risk evaluation of sediment metals in a tropical eutrophic bay, Guanabara Bay, Southeast Atlantic. *Mar Pollut Bull* 109:435–455
- Acquavita A, Emili A, Covelli S, Faganeli J, Predonzani S, Koron N, Carrasco L (2012) The effects of resuspension on the fate of Hg in contaminated sediments (Marano and Grado Lagoons, Italy): short-term simulation experiments. *Estuar Coast Shelf Sci* 113:32–40
- Aguiar VMC, Abuchaca PFF, Baptista Neto JA, Oliveira AS (2018) Environmental assessment concerning trace metals and ecological risks at Guanabara Bay, RJ. *Brazil Environ Monit Assess* 190:448. <https://doi.org/10.1007/s10661-018-6833-x>
- Aguiar VMC, Baptista-Neto JA, Fonseca EM (2022) Assessment of eutrophication through ecological indicators at the entrance of a tropical urbanized estuary. *J Integr Coast Zone Manag* 22(3):175–192. <https://doi.org/10.5894/jcoi-n379>

- Asad F, Batool N, Nadeem A, Bano S, Jamal R, Ali S (2024) Fe-NPs and Zn-NPs: advancing aquaculture performance through nanotechnology. *Biol Trace Elem Res* 202:2828–2842. <https://doi.org/10.1007/s12011-023-03850-6>
- Baptista-Neto JA, Peixoto TCS, Smith BJ, McAlister JJ, Patchineelam SM, Patchineelam SR, Fonseca EM (2013) Geochronology and heavy metal flux to Guanabara Bay, Rio de Janeiro state: a preliminary study. *An Acad Bras Cienc* 85(4):1317–1327
- Barreira J, Garnier J, Cunha B, Tonh   MS, Dantas E, Freyrier R, Machado W, Babiski M, Ara  jo R, Ara  jo DF (2025) Zinc stable isotopes in mangrove crabs as tracers of anthropogenic contamination in a tropical estuary. *Chemosphere* 382:144484. <https://doi.org/10.1016/j.chemosphere.2025.144484>
- Barrocas PRG, Wasserman JC (1998) Mercury behaviour in sediments from a sub-tropical coastal environment in SE Brazil. In: Wasserman JC, Silva-Filho EV, Villas-Boas R (eds) *Environmental geochemistry in the tropics*, vol vol 72. Springer, Heidelberg, p pp 271–284
- Bejarano-Ramirez I, Jurado JM, Mu  iz-Valencia R, Alc  zar A, Ceballos-Maga  a SG, Olivios-Ortiz A, Rangel O (2017) Comparative study of As, Cd, Cu, Cr, Mg, Mn, Ni, Pb and Zn concentrations between sediment and water from estuary and port. *Int J Environ Sci Technol* 14:1333–1342
- Brady JP, Kinaev I, Goonetilleke A, Ayoko GA (2016) Comparison of partial extraction reagents for assessing potential bioavailability of heavy metals in sediments. *Mar Pollut Bull* 106:329–344. <https://doi.org/10.1016/j.marpolbul.2016.03.029>
- Caetano M, Madureira MJ, Vale C (2003) Metal remobilization during resuspension of anoxic contaminated sediment: short-term laboratory study. *Water Air Soil Pollut* 143:23–40
- Champiaw P, Ponsin M, Numprasantai A (2023) Effects of sediment resuspension and changes in water nutrient concentrations on the remobilization of lead from contaminated sediments in Klity Creek. *Thailand J Environ Manage* 339:117909
- Chen C, Ju Y, Chen C, Dong C (2019) Changes in the total content and speciation patterns of metals in the dredged sediments after ocean dumping: Taiwan continental slope. *Ocean Coast Manag* 181:104893
- Choueri RB, Cesar A, Abessa DMS, Torres RJ, Morais RD, Riba I, Pereira CDS, Nascimento MRL, Mozeto AA, DelValls TA (2009) Development of site-specific sediment quality guidelines for North and South Atlantic littoral zones: comparison against national and international sediment quality benchmarks. *J Hazard Mater* 170:320–331. <https://doi.org/10.1016/j.jhazmat.2009.04.093>
- CONAMA (2012) Procedures for the evaluation of dredging materials in Brazilian jurisdictional waters. Resolution No. 454/2012. *Di  rio Oficial da Rep  blica Federativa do Brasil, Bras  lia* (in Portuguese)
- Corr  a JJM, Cutrim MVJ, da Cruz QS (2023) Evaluation of metal contamination in surface sediments and macroalgae in mangrove and port complex ecosystems on the Brazilian equatorial margin. *Environ Monit Assess* 195:432. <https://doi.org/10.1007/s10661-023-11024-z>
- d'Errico G, Nardi A, Benedetti M, Mezzelani M, Fattorini D, Di Carlo M, Regoli F (2021) Application of a multidisciplinary weight of evidence approach as a tool for monitoring the ecological risk of dredging activities. *Front Mar Sci* 8:765256
- Di Nanno MP, Curutchet G, Ratto S (2007) Anaerobic sediment potential acidification and metal release risk assessment by chemical characterization and batch resuspension experiments. *J Soils Sediments* 7:187–194. <https://doi.org/10.1065/jss2007.04.220>
- Fetters KJ, Costello DM, Hammerschmidt CR, Burton GA (2016) Toxicological effects of short-term resuspension of metal-contaminated freshwater and marine sediments. *Environ Toxicol Chem* 35(3):676–686
- Heiri O, Lotter AF, Lemcke G (2001) Loss on ignition as a method for estimating organic and carbonate content in sediments: reproducibility and comparability of results. *J Paleolimnol* 25:101–110
- Huerta-Diaz MA, Morse JW (1990) A quantitative method for determination of trace metal concentration in sedimentary pyrite. *Marine Chemistry* 29:119–144
- IBGE (2022) Censo Demogr  fico, Rio de Janeiro, Brazil. Instituto Brasileiro de Geografia e Estat  stica (IBGE), Rio de Janeiro
- Kjerfve B, Ribeiro CHA, Dias GTM, Filippo AM, Quaresma VS (1997) Oceanographic characteristics of an impacted coastal bay: Ba   de Guanabara, Rio de Janeiro, Brazil. *Cont Shelf Res* 17:1609–1643
- Larner BL, Palmer AS, Seen AJ, Townsend AT (2008) A comparison of an optimised sequential extraction procedure and dilute acid leaching of elements in anoxic sediments, including the effects of oxidation on sediment metal partitioning. *Anal Chim Acta* 608:147–157. <https://doi.org/10.1016/j.aca.2007.12.016>
- Lee DI, Eom KH, Kim GY, Baek GW (2010) Scoping the effective marine environmental assessment of dredging and ocean disposal of coastal sediments in Korea. *Mar Policy* 34(5):1082–1092. <https://doi.org/10.1016/j.marpol.2010.03.008>
- Leleyter L, Rousseau C, Biree L, Baraud F (2012) Comparison of EDTA, HCl and sequential extraction procedures, for selected metals (Cu, Mn, Pb, Zn), in soils, riverine and marine sediments. *J Geochem Explor* 116–117:51–59. <https://doi.org/10.1016/j.jgexplo.2012.03.006>
- Li P, Tian R, Liu R (2019) Solute geochemistry and multivariate analysis of water quality in the Guohua phosphorite mine, Guizhou Province, China. *Exposure and Health* 11(2):81–94. <https://doi.org/10.1007/s12403-018-0277-y>
- Linkov I, Loney D, Cormier S, Satterstrom FK, Bridges T (2009) Weight-of-evidence evaluation in environmental assessment: review of qualitative and quantitative approaches. *Sci Total Environ* 407(19):5199–5205
- Long ER, MacDonald DD, Smith SL, Calder FD (1995) Incidence of adverse biological effects within ranges of chemical concentrations in marine and estuarine sediments. *Environ Manage* 19:81–97
- Machado W, Rodrigues APC, Bidone ED, Sella SM, Santelli RE (2011) Evaluation of Cu potential bioavailability changes upon coastal sediment resuspension: an example on how to improve the assessment of sediment dredging environmental risks. *Environ Sci Pollut Res* 18:1033–1036
- Maddock JEI, Carvalho MF, Santelli RE, Machado W (2007) Contaminant metal behaviour during re-suspension of sulphidic estuarine sediments. *Water Air Soil Pollut* 181:193–200
- McAlister JJ, Smith BJ, Curran JA (2010) The use of sequential extraction to examine iron and trace metal mobilisation and the case-hardening of building sandstone: a preliminary investigation. *Microchemical J* 74(1):5–18. [https://doi.org/10.1016/S0026-265X\(02\)00043-7](https://doi.org/10.1016/S0026-265X(02)00043-7)
- Monte CN, Rodrigues APC, Cordeiro RC, Freire AS, Santelli RE, Machado W (2015) Changes in Cd and Zn bioavailability upon an experimental resuspension of highly contaminated coastal sediments from a tropical estuary. *Sustain Water Resour Manag* 1:332–335
- Monte CN, Rodrigues APC, Silva MC, Ferreira LJS, Monte G, Silveira CS, Cordeiro RC, Machado W (2023) Assessment of eutrophication from phosphorus remobilization after resuspension of coastal sediments from an urban tropical estuary. *Environ Sci Pollut Res* 30:65500–65511
- Monte CN, Rodrigues APC, Campos BG, Moreira LB, Grebbe B, Pauly GFE, Cruz ACF, Silva LID, Cordeiro RC, Abessa DMS, Machado W (2025) Influence of seasonality on the toxicity and metals bioavailability after the resuspension of sediments from a tropical estuary. *Estuar Coast Shelf Sci* 315:109167. <https://doi.org/10.1016/j.ecss.2025.109167>
- Moreira LB, Castro IB, Filmann G, Peres TF, Belmino IKC, Sasaki ST, Taniguchi S, Bicego MC, Marins RV, Lacerda LD, Costa-Lotufo LV, Abessa DMS (2021) Dredging impacts on the toxicity and development of sediment quality values in a semi-arid region (Cear   state, NE Brazil). *Environ Res* 193:110595. <https://doi.org/10.1016/j.envres.2020.110525>
- Morelle J, Clauquin P, Orvain F (2020) Evidence for better microphytobenthos dynamics in mixed sand/mud zones than in pure sand or mud intertidal flats (Seine estuary, Normandy, France). *PLoS ONE* 15(8):e0237211
- Morse JW (1994) Interactions of trace metals with authigenic sulfide minerals: implications for their bioavailability. *Mar Chem* 46:1–6
-   z  ker K, Er  z C, Terzi Y (2022) Evaluation of toxic metals in different grain size fractions of sediments of the southeastern Black Sea. *Mar Pollut Bull*. <https://doi.org/10.1016/j.marpolbul.2022.113959>
-   z  ker K, & Terzi, Y. (2025). Seasonal Geochemical Analysis of Toxic Metal Contamination and Ecological Risk Assessment in Sediments of the De  irmendere River Basin and Estuary, T  rkiye. *Water, Air, and Soil Pollution*, 236(8). <https://doi.org/10.1007/s11270-025-08191-9>
- Pe  a-Icart M, Mendiguch  a C, Villanueva-Tagle ME, Pomares-Alfonso MS, Moreno C (2014) Revisiting methods for the determination of bioavailable metals in coastal sediments. *Mar Pollut Bull* 89:67–74
- Puig M, Azarkamand S, Wooldridge C, Sel  n V, Darbra RM (2022) Insights on the environmental management system of the European port sector. *Sci Total Environ* 806:150550. <https://doi.org/10.1016/j.scitotenv.2021.150550>
- Rauret G (1998) Extraction procedures for the determination of heavy metals in contaminated soil and sediment. *Talanta* 46:449–455
- Reid J, Seiler L, Siegle E (2022) The influence of dredging on estuarine hydrodynamics: historical evolution of the Santos estuarine system, Brazil. *Estuar Coast Shelf Sci* 279:108131. <https://doi.org/10.1016/j.ecss.2022.108131>
- Riza M, Ehsan MN, Pervez MN, Khyum MMO, Cai Y, Naddeo V (2023) Control of eutrophication in aquatic ecosystems by sustainable dredging: effectiveness,

- environmental impacts, and implications. *Case Stud Chem Environ Eng* 7:100297. <https://doi.org/10.1016/j.csee.2023.100297>
- Rodrigues APC, Lemos AP, Monte CN, Rodrigues SK, Cesar RG, Machado W (2017) Environmental risk in a coastal zone from Rio de Janeiro state (Brazil) due to dredging activities. In: Araújo CVM, Shinn C (eds) *Ecotoxicology in Latin America*. Nova Publishers, New York, pp 183–200
- Rodrigues APC, Monte CN, Figueira G, Freitas AR, Santelli RE, Braz BF, Machado W (2024) Metal mobility after resuspension of contaminated sediments from a tropical urban bay. *Rev Bras Geogr Física* 17(2):994–1014
- Safhi AM, Mejjad N, El Fadil IH, Bortali M (2024) Dredged materials in Morocco: current practices, policies, and roadmap for sustainable management. *Case Stud Constr Mater* 20:e03045. <https://doi.org/10.1016/j.cscm.2024.e03045>
- Seixas Filho JT, Mello SCRP, Faria AS, Souza LL (2020) Análise socioambiental da poluição por esgoto da Baía de Guanabara do Rio de Janeiro. *Rev Valore e-5022*
- Shao S, Liu H, Tai X, Zheng F, Li J, Li Y (2020) Speciation and migration of heavy metals in sediment cores of urban wetland: bioavailability and risks. *Environ Sci Pollut Res* 27:23914–23925. <https://doi.org/10.1007/s11356-020-08719-y>
- Silva DA, de Souza Fraga M, Lyra GB, Cecílio RA, Pereira CR, Cunha-Zeri G, Zeri M, Abreu MC (2024) Assessment of water quality and identification of priority areas for intervention in Guanabara Bay basin, Rio de Janeiro, Brazil, using nonparametric and multivariate statistical methods. *Environ Monit Assess* 196:912. <https://doi.org/10.1007/s10661-024-13002-5>
- Silveira RP, Rodrigues APC, Santelli RE, Cordeiro RC, Bidone ED (2011) Mass balance in the monitoring of pollutants in tidal rivers of the Guanabara Bay, Rio de Janeiro, Brazil. *Environ Monit Assess* 181:165–173
- Soetan O, Zhu Q, Feng H (2025) Spatiotemporal evaluation of water quality, metal pollution, and human health risks in a dredged Urban River, New Jersey, USA. *Environ Geochem Health* 47:283. <https://doi.org/10.1007/s10653-025-02579-6>
- Sumi T, Hirose T (2009) Accumulation of sediment in reservoirs. Water storage, transport and distribution. UNESCO-IHE and EOLSS Publishers Co. Ltd., Paris, pp 224–252
- Sutherland RA (2010) BCR⁺-701: A review of 10-years of sequential extraction analyses. *Analytica Chimica Acta* 680(1–2):10–20. <https://doi.org/10.1016/j.aca.2010.09.016>
- Valentin JL, Tenenbaum DR, Bonecker AC, Bonecker SLC, Nogueira C, Villac MC (1999) O Sistema Planctônico da Baía de Guanabara: Síntese do Conhecimento. *Ecologia dos Ambientes Costeiros do Estado do Rio de Janeiro. Series Oecologia Brasiliensis*, Rio de Janeiro, PPGE-UFRJ, VII:35–39
- Vicente MC, Trevisan CL, Carvalho ACB, Oliveira BCV, Rezende CE, Machado WV, Wasserman JC (2024) Geochemical fractionation of trace metals and ecological risk assessment of surface sediments in Sepetiba Bay, Brazil. *Environ Sci Pollut Res Int*. 31(9):14254–14269. <https://doi.org/10.1007/s11356-024-32095-6>
- Wasserman JC, Wasserman MA, Barrocas PRG, Almeida AM (2016) Predicting pollutant concentrations in the water column during dredging operations: Implications for sediment quality criteria. *Mar Pollut Bull* 108:24–32. <https://doi.org/10.1016/j.marpolbul.2016.05.005>
- Wu G, Reichart G-J, Kraal P (2025) Reactivity and potential mobility of metals in human-impacted harbor sediments (Port of Rotterdam, the Netherlands). *J Soils Sediments* 25:277–291. <https://doi.org/10.1007/s11368-024-03936-1>
- Xie M, Alsina MA, Yuen J, Packman AI, Gaillard J (2019) Effects of resuspension on the mobility and chemical speciation of zinc in contaminated sediments. *J Hazard Mater* 364(15):300–308
- Yan J, Li F (2023) Effects of sediment dredging on freshwater system: a comprehensive review. *Environ Sci Pollut Res* 30:119612–119626. <https://doi.org/10.1007/s11356-023-30851-8>
- Yang J, Chen L, Liu LZ, Shi WL, Meng XZ (2014) Comprehensive risk assessment of heavy metals in lake sediment from public parks in Shanghai. *Ecotoxicol Environ Saf* 102:129–135. <https://doi.org/10.1016/j.ecoenv.2014.01.010>
- Yu JH, Chen QW, Zhang JY, Zhong JC, Fan CX, Hu LM, Shi WQ, Yu WY, Zhang YL (2019) In situ simulation of thin-layer dredging effects on sediment metal release across the sediment-water interface. *Sci Total Environ* 658:501–509
- Yu Z, Liu E, Lin Q, Zhang E, Yang F, Wei C, Shen J (2021) Comprehensive assessment of heavy metal pollution and ecological risk in lake sediment by combining total concentration and chemical partitioning. *Environ Pollut* 269. <https://doi.org/10.1016/j.envpol.2020.116212>
- Zhong XL, Zhou SL, Zhu Q, Zhao QG (2011) Fraction distribution and bioavailability of soil heavy metals in the Yangtze River Delta—a case study of Kunshan City in Jiangsu Province, China. *J Hazard Mater* 198:13–21